SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	12-10-2026	**Duration:**	60 minutes

Code of Conduct

I VIGNESHWARAN R / 1911260109 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Problem Statement: Library Management System for a College Library

1. Understand the Problem Statement

1.1 Problem in my own words

A college library holds thousands of books, journals and e-resources used by students, faculty and staff. Today, book issue/return, member records and fine collection are handled with registers and spreadsheets. This makes searching slow, causes errors in records, loses books, and leaves librarians without reliable data. The problem is to build a computerised Library Management System (LMS) that automates these activities and provides accurate, real-time information to every user.

1.2 Objectives of the proposed system

- Automate cataloguing, issue, return, renewal and reservation of books.
- Provide fast online search of the catalogue by title, author, ISBN or subject.
- Calculate fines automatically and send due-date reminders by email/SMS.
- Maintain accurate member and inventory records with a complete audit trail.
- Generate reports (most-borrowed books, overdue list, stock status) for decision making.
- Provide secure, role-based access for students, faculty and librarians.

1.3 Scope

In scope: member management, catalogue management, circulation (issue/return/renew/reserve), fine management, notifications, reports, and a web interface (mobile-friendly). **Out of scope:** payment-gateway integration beyond basic fine recording, RFID hardware, and inter-library loan in the first release.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders, roles and responsibilities

Stakeholder	Role and Responsibility
Students	Search books, reserve, renew, view dues and borrowing history; return books on time.
Faculty	Same as students, with a longer loan period; recommend new books for purchase.
Librarian	Add/update books, issue and return books, manage members, collect fines, generate reports.
Library Assistants	Handle day-to-day counter work, shelving and stock verification.
System Administrator	Manage user accounts, roles, backups, security and system maintenance.
College Management / HOD	Approve budget and policies, review usage reports, decide on procurement.
Development Team	Analyse, design, develop, test and deploy the system; provide support.
Accounts Department	Verify fine collections and reconcile receipts.

2.2 Major challenges and issues

- **Students/Faculty:** long queues, no way to check book availability remotely, forgotten due dates leading to fines.
- **Librarian/Staff:** heavy manual workload, duplicate entries, difficulty tracing lost or misplaced books.
- **Management:** no consolidated statistics, difficulty planning purchases and budgets.
- **Admin/Developers:** data migration from old records, security of personal data, user training, limited budget.

3. Analyze the Current Approach

3.1 Existing system / current process

Members fill a slip or sign a register at the counter. The librarian searches the shelf or an Excel sheet, writes the issue date in the register and a due-date card in the book. On return, the librarian manually checks the date, calculates the fine on paper and updates the register. Reports are prepared by hand at the end of the semester.

3.2 Strengths and limitations

Strengths	Limitations
Simple and needs no technical skill. Low initial cost. Works without internet or power.	Slow and time consuming; long queues. High risk of human error and data loss. No remote search or reservation. Fine calculation is inconsistent. No reports or analytics. Poor security and no audit trail.

3.3 Gaps to be addressed

- No centralised, real-time database of books and members.
- No automated reminders, fine calculation or reservation.
- No role-based access control or backup mechanism.
- No reporting and analytics for decision making.
- System cannot scale as the number of books and users grows.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed software-based solution

A web-based Library Management System with a three-tier architecture: **Presentation layer** (responsive web/mobile UI), **Business logic layer** (REST API services) and **Data layer** (relational database such as MySQL/PostgreSQL). Users log in with role-based accounts; all transactions are stored centrally and processed in real time.

4.2 Major functional modules

Module	Description
User & Role Management	Registration, login, roles (student, faculty, librarian, admin), profile management.
Catalogue Management	Add/edit/delete books, categories, copies, ISBN, shelf location.
Search Module	Keyword and filter-based search with availability status.
Circulation Module	Issue, return, renew and reserve books; barcode/ID based lookup.
Fine Management	Auto-calculate overdue fines, record payments, generate receipts.
Notification Module	Email/SMS alerts for due dates, reservations and new arrivals.
Reports & Analytics	Overdue list, popular books, member activity, stock report (PDF/Excel).
Admin & Backup	System settings, audit logs, scheduled backup and restore.

4.3 Requirements

Functional Requirements	Non-Functional Requirements
FR1: Secure login with role-based access. FR2: Add, update and delete book records. FR3: Search books by title/author/ISBN/subject. FR4: Issue, return, renew and reserve books. FR5: Auto-calculate fines for overdue books. FR6: Send due-date and reservation notifications. FR7: Generate reports. FR8: Maintain issue history for each member.	**Performance:** search result within 2 seconds. **Scalability:** support 5,000+ users and 1 lakh+ records. **Availability:** 99% uptime during college hours. **Security:** encrypted passwords, HTTPS, access control. **Usability:** simple, responsive interface; minimal training. **Reliability:** daily automatic backup. **Maintainability:** modular, documented code.

4.4 Software engineering principles incorporated

Principle	How it is incorporated
Modularity	System is split into independent modules (catalogue, circulation, fines, etc.) with clear interfaces, so each can be developed and tested separately.
Scalability	Three-tier architecture, indexed database and stateless REST APIs allow adding servers or users without redesign.
Maintainability	Clean coding standards, layered design, version control (Git) and documentation make bug fixes and changes easy.
Reliability	Input validation, transaction handling, exception handling, automated testing and scheduled backups.
Usability	Simple navigation, responsive design, clear messages and role-specific dashboards.
Security	Hashed passwords, role-based access control, HTTPS, SQL-injection protection and audit logs.

5. Justify the Chosen Methodology / Framework

5.1 Selected model: Agile – Scrum

Work is done in 2-week sprints with a Product Owner (Head Librarian), Scrum Master and Development Team. A prioritised product backlog (e.g. login, catalogue, issue/return, fines, reports) is delivered incrementally, with a sprint review and retrospective at the end of each sprint.

5.2 Why Scrum is suitable

- **Changing requirements:** librarians and students will give new ideas after seeing early versions; Scrum handles change easily, whereas Waterfall does not.
- **Early value:** core features (catalogue and issue/return) can be released first and used quickly.
- **Continuous stakeholder feedback** through sprint reviews reduces the risk of building the wrong system.
- **Risk control:** small increments make problems visible early and cheap to fix.
- **Team transparency:** daily stand-ups, sprint board and burndown charts track progress.

5.3 Support for development and maintenance

Each sprint includes design, coding, testing and demo, so quality is built in. Continuous integration, automated tests and documentation make later maintenance easy. After deployment, new requirements simply enter the backlog and are delivered in later sprints.

6. Discuss Expected Outcomes and Limitations

6.1 Expected benefits

- Issue/return time reduced from minutes to seconds; queues eliminated.
- Accurate, real-time records and fewer lost books.
- 24x7 online search, renewal and reservation for users.
- Automatic fines and reminders reduce disputes and late returns.
- Management gets reports for better purchasing decisions.

6.2 How challenges are addressed

Challenge	Solution
Long queues and manual work	Automated circulation with barcode/ID lookup.
No remote availability check	Online catalogue search and reservation.
Forgotten due dates	Automatic email/SMS reminders.
Errors and lost data	Central database with validation and daily backup.
No reports	Reports and analytics module.

6.3 Risks, implementation challenges and limitations

- **Risks:** data-migration errors, user resistance to change, power/internet failure, security threats.
- **Mitigation:** pilot run, staff training, offline-tolerant design, regular backups, security testing.
- **Limitations:** depends on internet and server availability; initial cost and training effort; no RFID or inter-library loan in release 1.

6.4 Future enhancements

- Mobile app with barcode/QR scanning.
- RFID-based self-checkout and anti-theft system.

- AI-based book recommendations.
- E-book and digital resource integration.
- Online fine payment gateway and inter-library loan.
- Integration with the college ERP / student information system.